

Bit Error Rate Analysis and Minimisation for Quantum Backscatter Communications under Practical Impairments

Corey Anderson, Deepak Mishra

School of Electrical Engineering and Telecommunications, University of New South Wales, Sydney, NSW 2052, Australia
Emails: corey.anderson@student.unsw.edu.au, d.mishra@unsw.edu.au

Abstract—Quantum backscatter communication (QBC) enhances passive sensing and ultra-low-power communication by exploiting the joint detection of entangled signal-idler photon pairs. Existing QBC literature primarily models the channel as a thermal bath. However, this idealised model neglects several physical impairments that arise in realistic polarisation-based systems. The contribution of this work is realistic BER characterisation beyond the standard thermal-bath assumption by incorporating three additional impairments: rotational jitter, temporal jitter and pointing jitter. Using Stokes vector formalism, Gaussian pulse-overlap modelling and an angular jitter model, analytical expressions are derived and their validity is confirmed through numerical simulation. In addition, an optimisation framework is developed for the combined pointing, rotation, and temporal jitter model, where pulse width and signal brightness are jointly designed using the data-rate and resource constraints. The results show that practical impairments introduce nonlinear degradation effects that are not captured by ideal thermal-bath models, while the optimisation reveals a non-trivial tradeoff between pulse width and signal brightness. Additionally, the combined pointing, rotation and temporal jitter model is used to show that rotation and pointing jitter dominate in the considered baseline regime and the robust operating regions for design parameters are identified. These findings provide a more realistic characterisation of polarisation-encoded QBC channels and offer useful engineering design guidance for practical system implementation.

I. INTRODUCTION

Quantum backscatter communication (QBC) is an emerging approach for ultra-low-power sensing and data transfer that leverages reflected signals and quantum correlations to improve reliability in noisy channels through the use of quantum illumination [1], [2]. While existing models suggest substantial performance gains using an idealised thermal-bath model, real systems will face more severe practical impairments that can weaken the distinguishing features exploited by the quantum receivers. Quantifying how these imperfections translate into bit error probability P_{err} is critical in bridging the gap between theory and practical deployment. This work addresses this need by analysing the bit error rate behaviour of quantum backscatter communication systems under multiple practical impairment models, as well as optimising the system for robust performance.

A. Motivation and Contributions

The current QBC literature primarily assumes the communication link channel is represented by an idealised thermal-

bath to characterise performance. While this model provides simple impairment considerations, it neglects several physically important impairments that arise in practical polarisation-based systems. This causes the idealised assumptions to break down, including polarisation rotation, temporal misalignment between signal and idler, and pointing errors. These effects directly reduce the distinguishability between quantum states and can significantly increase P_{err} in realistic operating conditions. As QBC moves toward experimentally relevant implementations, incorporating such impairments into the system model is critical for accurately characterising performance limits and understanding how practical constraints impact the achievable quantum advantage.

The main contributions of this work are summarised as follows:

- Development of rotation, temporal, and pointing jitter within a polarisation-encoded framework, and the combination of all three impairments into a unified model.
- Derivation of closed-form analytical expressions for P_{err} under each impairment model using physically motivated approximations.
- Rigorous convergence analysis for the derived analytical series expressions, including proofs of absolute convergence and practical truncation-length rules for accurate numerical evaluation.
- Validation of the analytical results through numerical simulations, with design insights extracted using sensitivity-based impairment comparisons.
- Formulation and solution of a constrained optimisation problem to minimise P_{err} with respect to key system parameters, providing information on fundamental design trade-offs.

B. Research Objectives

This work addresses the following research questions:

- How do practical impairments such as rotation, temporal and pointing errors degrade the effective distinguishability of quantum states in QBC systems?
- How does each impairment individually and jointly affect the behaviour of P_{err} compared to the ideal thermal-bath model?
- Under what operating regimes do these impairments become dominant, and what operating conditions should

these systems be engineered to operate at to ensure robust behaviour?

- How accurately do the analytical models capture these impairments compared to numerical evaluation?
- What system parameter choices minimise P_{err} under practical constraints?

II. STATE-OF-THE-ART

A. Classical Backscattering

Classical backscatter communication (CBC) provides the foundational paradigm upon which QBC systems are built. Unlike conventional wireless systems that actively generate RF signals, backscatter systems convey information by modulating and reflecting an incident waveform from an external source, eliminating the need for an onboard RF transmitter [3], [4]. A canonical system consists of a signal source, a backscatter tag, and a receiver. The tag encodes information by impedance switching, thereby modifying the reflection coefficient and imprinting data onto the incident signal [3]. This enables ultra-low-power operation, making backscatter communications suitable for IoT-scale deployments. CBC systems can broadly be categorised into

- **Monostatic:** co-located transmitter and receiver
- **Bistatic:** separated source and receiver, improving the geometry and range

Recent developments include ambient backscatter, where existing RF signals (e.g., from WiFi, TV) act as carriers, enabling battery-free communication without dedicated transmitters [4]. Despite these advantages, CBC systems still suffer from weak reflected signals buried in noise and interference, and poor detection performance in low-SNR environments. These limitations motivate QBC approaches, which enhance signal detection in low-SNR regimes using entanglement-assisted receivers.

B. Quantum Illumination

Lloyd introduced the concept of quantum illumination (QI) for enhanced photodetection using entangled photon pairs [1]. In this seminal work, signal-idler states in the low-brightness regime were proposed for target detection in noisy environments, demonstrating a theoretical sensitivity advantage over classical illumination schemes. However, the model relied on single-photon entanglement and extremely low signal flux, limiting its practical feasibility for realistic communication systems.

Tan *et al.* extended Lloyd's single-photon model using two-mode squeezed vacuum (TMSV) states [2]. They showed that even if entanglement is destroyed at the receiver, the TMSV input introduces phase-sensitive correlations that exceed the maximum allowed for any classical state with the same energy. They used optimal quantum reception to demonstrate a 6 dB improvement in the error probability exponent compared to optimal classical coherent-state transmitters. However, these results are derived under idealised thermal-bath channel assumptions and do not account for practical impairments that degrade signal-idler correlations in realistic systems.

C. Quantum Radar

Quantum radar is a broader class of sensing systems that use quantum states of light to enhance detection performance in noisy environments. Many practical implementations are based on quantum illumination, which exploits non-classical correlations to improve target detection and estimation beyond classical limits. Assouly *et al.* [5] provide one of the first clear experimental demonstrations of a microwave quantum advantage. Using a superconducting circuit, they achieved more than 20% better performance than the best possible classical radar. They noted that the main limiting factor is the purity of the initial entangled state, as this is crucial for quantum-correlated detection at the receiver.

In contrast, Jonsson and Ankel [6] provide a systems-level evaluation of quantum radar, arguing that even if a theoretical quantum advantage exists, practical usefulness in conventional radar settings is severely limited by the low-brightness requirement and short range. Additionally, they note that current practical receivers often achieve only a 3 dB gain over classical receivers, whereas the ideal bound is 6 dB, which highlights the need for practical impairment modelling. These experimental and theoretical results indicate that while quantum advantage is achievable, it is highly sensitive to non-idealities such as idler loss, noise calibration, and entangled-state impurity.

D. QBC Protocols and Receivers

Di Candia *et al.* extended the quantum illumination paradigm to communication applications by proposing a QBC protocol [7]. In this framework, entangled signal-idler photon pairs are generated at the transmitter, with the signal interacting with the tag for modulation and the idler retained at the receiver for joint measurement. By mapping communication symbols to quantum states and performing joint detection, the system inherits the error-exponent advantage of quantum illumination, typically characterised via the quantum Chernoff bound. It is shown that QBC achieves up to a 6 dB improvement in the error probability exponent for pulse-amplitude modulation (PAM) and binary phase-shift keying (BPSK), and a 3 dB gain for quadrature phase-shift keying (QPSK), relative to classical backscatter systems.

Khalifa *et al.* [8] propose an alternative QBC system which uses photon-number (Fock) states. Their results demonstrate that an error-exponent advantage is achievable in high-noise environments, but disappears in high-SNR regimes. Furthermore, the receiver performance is highly sensitive to timing mismatch, requiring precise delay synchronisation between signal and idler, which limits effectiveness in practical environments.

Arnon [9] extends channel impairment modelling by introducing a pointing-jitter model arising from antenna misalignment in free-space backscatter systems. By modelling angular misalignment as a random Rayleigh process and performing numerical analysis, it is shown that pointing errors lead to an exponential decay in detection performance at the receiver.

Despite these advances, most existing QBC models rely on idealised thermal-bath channel assumptions and do not incorporate stochastic impairments that degrade signal-idler correlations in realistic environments. While some works consider specific effects such as timing mismatch or pointing errors, there is no unified analytical framework that jointly models multiple practical impairments and quantifies their impact on error probability. This motivates the need for a BER analysis framework for QBC systems under realistic channel conditions. The predominant thermal-bath model assumes that the only channel impairment is the additive photon noise, neglecting impairment mechanisms that directly reduce signal-idler correlations. For polarisation-encoded systems, impairments such as random polarisation rotation, temporal misalignment, and pointing errors act multiplicatively on the effective distinguishability between quantum states rather than additively on noise power. Thus, these effects are not accurately represented within the standard thermal-bath framework and require explicit modelling to capture their effect on BER performance.

E. Polarisation Modulations

Polarisation modulation provides a natural encoding framework for backscatter systems and serves as a useful classical baseline for comparison with quantum-enhanced schemes. Polarisation-based backscatter modulation schemes have been proposed by Zeng *et al.* [10], which employ binary and differential polarisation shift keying (BPolSK and DPolSK), where information is encoded in orthogonal polarisation states and detected using a dual-polarised receiver. A major contribution is the derivation of closed-form BER expressions, showing exponential error decay for BPolSK, while DPolSK enables non-coherent detection and is more robust to polarisation mismatches.

However, the analysis is restricted to classical detection models and deterministic mismatch effects, without incorporating stochastic impairments or quantum-enhanced joint detection mechanisms. There is currently a lack of analytical work that considers stochastic impairment effects for polarisation modulation in QBC systems, such as random polarisation drift of the signal mode and its impact on BER.

III. SYSTEM DESCRIPTION

A. Network Topology and Access Protocol

Following [7], we adopt a bistatic QBC link in which the transmitter (TX) emits pairs of entangled photons; the signal photon and the idler photon (S and I) as shown in Figure 1. The S-photon interacts with a passive tag that modulates the polarisation, and a receiver (RX) performs a joint measurement on the backscattered S-photon alongside the I-photon.

The TX generates M identical S-I mode pairs. Each pair is prepared in a two-mode squeezed vacuum (TMSV) state given by

$$|\psi\rangle_{SI} = \sum_{n=0}^{\infty} \sqrt{\frac{N_S^n}{(N_S + 1)^{n+1}}} |n\rangle_S |n\rangle_I \quad (1)$$

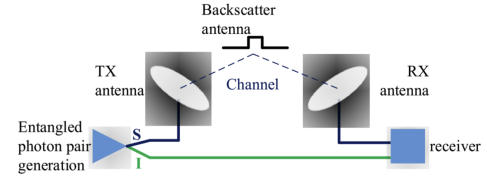


Fig. 1: A model of a bistatic QBC system [7]

where N_S is the average number of photons in each of the signal and idler modes (signal brightness) [7]. The I-mode is sent to the receiver assuming no loss. Meanwhile, the S-mode is sent to the backscatter tag. We assume no noise in the TX to tag stage, only in the return stage to match the standard QBC return-link-dominated noise assumption.

The RX performs a joint quantum measurement on each returned signal mode and its corresponding idler. As in [7], this is implemented through a parametric amplifier or a sum-frequency-generation receiver.

B. Wireless Channel Model

After scattering from the tag, the S-photon propagates through a noisy channel. In standard QBC, this channel is modelled as a low-reflectivity beamsplitter that mixes the signal with the thermal state [7]. This is shown in Figure 2, and given by

$$\hat{a}_R = \sqrt{\eta}e^{-i\phi}\hat{a}_S + \sqrt{1-\eta}\hat{a}_Z \quad (2)$$

where $\hat{a}_R$, $\hat{a}_S$, $\hat{a}_Z$ correspond to the received, signal and noise modes, and η is the round trip transmissivity (with $\eta \ll 1$ for the high noise environment). Unlike the phase-modulated setting in [7], the global phase shift ϕ introduced by the beamsplitter is irrelevant for polarisation modulation and can therefore be omitted with $\phi = 0$.

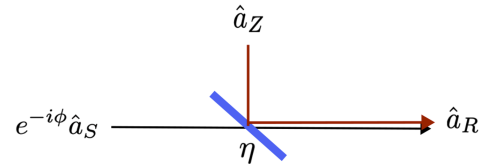


Fig. 2: Beamsplitter model for the thermal bath channel. $e^{-i\phi}\hat{a}_S$ is the modulated signal mode travelling to the receiver, $\hat{a}_Z$ is the thermal noise mode that is mixed with the signal through a beamsplitter with transmissivity η . $\hat{a}_R$ is the received mode.

C. Backscatter Tag Modulation

The tag performs polarisation modulation on the incident signal photon before backscattering. The Stokes vector formalism covered in [11] is used to represent the polarisation state of the signal photon. Consider the Stokes vector for a

linearly polarised light at an angle θ counterclockwise from the x -axis:

$$\mathbf{S} = \begin{pmatrix} |E_x|^2 - |E_y|^2 \\ 2 \operatorname{Re}(E_x^* E_y) \\ -2 \operatorname{Im}(E_x^* E_y) \end{pmatrix} = \begin{pmatrix} \cos^2 \theta - \sin^2 \theta \\ 2 \cos \theta \sin \theta \\ 0 \end{pmatrix} = \begin{pmatrix} \cos 2\theta \\ \sin 2\theta \\ 0 \end{pmatrix} \quad (3)$$

The tag modulates the bits onto the signal photon using two orthogonal states. The horizontal polarisation $\mathbf{S}_H$ corresponds to $\theta = 0$ and the vertical polarisation $\mathbf{S}_V$ corresponds to $\theta = \frac{\pi}{2}$. The two orthogonal Stokes vectors are therefore

$$\mathbf{S}_H = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{S}_V = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}. \quad (4)$$

At the receiver, the polarisation state is estimated by projecting the received Stokes vector onto the horizontal polarisation axis using the dot product

$$\mathbf{S}_{\text{rx}} = \begin{cases} \mathbf{S}_H, & \text{if } \mathbf{S}_{\text{rx}} \cdot \mathbf{S}_H > 0 \\ \mathbf{S}_V, & \text{if } \mathbf{S}_{\text{rx}} \cdot \mathbf{S}_H < 0. \end{cases} \quad (5)$$

In section IV-B2, this detection framework under the rotation jitter impairment will be considered.

IV. PROBLEM DEFINITION

A. Chernoff-bound Performance Metric

In order to evaluate the performance of the proposed modulation scheme, the bit error probability P_{err} is adopted as the metric. P_{err} can be tightly bounded using the quantum Chernoff bound as used in [7]:

$$P_{\text{err}} \leq \exp \left(-\frac{d^2 N_S M}{k N_Z} \right) \quad (6)$$

where M is the number of mode pairs, k is a constant determined by the receiver architecture, and N_Z is the thermal background noise photon number. The d in the exponent is the effective distinguishability parameter such that increasing d results in lower P_{err} . For a classical receiver, d is simply the Euclidean distance between the constellation points, arising from heterodyne detection [7]. In the quantum receiver, d represents the distinguishability between quantum states under joint signal-idler measurement, which incorporates quantum correlations rather than merely geometric separation. Note that in this case, P_{err} depends on distinguishability d , not SNR alone, which is a key distinction from classical backscatter systems. An exact expression for P_{err} is generally difficult to obtain for quantum state discrimination problems, particularly when further practical impairments are included. The quantum Chernoff bound is therefore adopted because it provides a tractable and asymptotically tight representation of error performance [12], while preserving the correct dependence on state distinguishability d .

Each independent impairment acts as a multiplicative reduction on the effective distinguishability d appearing in the

Chernoff bound. This is because the error exponent depends on the signal-idler correlation amplitude, and each impairment is assumed to independently attenuate this correlation through a scaling term K_n . Therefore, under independence, these effects combine multiplicatively. In quantum illumination, channel loss is known to scale the signal-idler correlation amplitude through the transmissivity [2], which supports modelling impairments as attenuation factors on the same underlying quantity. If each impairment K_n independently attenuates the correlation, sequential attenuation results in a product:

$$d \longrightarrow d K_1 K_2 K_3 \dots,$$

where each factor $0 < K_n < 1$ represents the contraction caused by the n th impairment. Physically, these impairments reduce the distinguishability of the polarisation-encoded quantum states, decreasing the receiver's ability to separate them.

For the rest of Section IV, signal models are introduced for each physical impairment, and the Chernoff bound expression for the corresponding impairments will be obtained as an expectation value. These impairments are:

1) Polarisation rotation

The propagation medium and antenna misalignment introduce a polarisation rotation and distorts the Stokes vector of the received signal, reducing the distinguishability between the two polarisation-encoded symbols.

2) Temporal Jitter

The signal and idler pulses experience an arrival-time mismatch that weakens the correlation exploited by the quantum receiver.

3) Pointing Jitter

Due to misalignment or vibration of the TX and/or RX, the signal isn't perfectly captured by the receiver. This effect has been studied in [9] but a closed-form expression will be derived and validated here.

In Section V, each of these impairment expressions will be solved to give an analytical bound for P_{err} . The purpose of deriving closed-form expressions is to enable rapid evaluation of performance across large parameter spaces. This allows for efficient optimisation of system parameters, facilitates sensitivity analysis to identify dominant impairments, and supports adaptive system tuning under changing channel conditions. Additionally, relying only on Monte Carlo methods obscures the insights into how impairments affect performance, whereas analytical expressions reveal explicit functional dependencies which engineering designs can use.

B. Practical Impairment Models

1) *Pointing Jitter Model*: Pointing errors in real systems can arise due to platform motion, beam steering instability, mechanical vibration of antennas, and tracking system errors. A channel model that incorporates the pointing direction error θ for the transmission of the signal photon is proposed in [9]. That model considers a free-space optical (FSO) model

where θ is a random variable of the Rayleigh distribution with probability density function $f(\theta) = \frac{\theta}{\sigma_p^2} e^{-\frac{\theta^2}{2\sigma_p^2}}$. The probability of bit error is modelled using the pointing loss factor $L(\theta) = \exp(-G\theta^2)$ to be

$$P_{\text{err}} \leq \alpha \int_0^\infty \frac{\theta}{\sigma_p^2} \exp\left(-\frac{\theta^2}{2\sigma_p^2} - k_1 G^2 e^{-2G\theta^2}\right) d\theta, \quad (7)$$

where G is the antenna gain and k_1 is proportional to the SNR.

2) *Rotation Error*: Rotation errors can arise in real setups due to atmospheric turbulence causing random birefringence, mechanical vibration of optical components and imperfect polarisation control at the tag. To model the effect, suppose the signal photon suffers some rotation error $\theta = \theta_0 + \epsilon$, where θ_0 was the original polarisation angle and ϵ is the rotation error. The impaired Stokes vector in equation 3 is now

$$\mathbf{S}_\epsilon = \begin{pmatrix} \cos(2\theta_0 + 2\epsilon) \\ \sin(2\theta_0 + 2\epsilon) \\ 0 \end{pmatrix}. \quad (8)$$

The detection operation in equation 5 gives

$$\begin{aligned} \mathbf{S}_\epsilon \cdot \mathbf{S}_H &= \cos(2\theta_0 + 2\epsilon) \cdot 1 + \sin(2\theta_0 + 2\epsilon) \cdot 0 + 0 \cdot 0 \\ &= \cos(2\theta_0) \cos(2\epsilon) - \sin(2\theta_0) \sin(2\epsilon). \end{aligned} \quad (9)$$

Thus, the detection method gives

$$\mathbf{S}_\epsilon \cdot \mathbf{S}_H = \begin{cases} \cos(2\epsilon), & \text{for horizontal polarisation } (\theta_0 = 0) \\ -\cos(2\epsilon), & \text{for vertical polarisation } (\theta_0 = \frac{\pi}{2}). \end{cases} \quad (11)$$

Comparing with the ideal case

$$\mathbf{S}_{\epsilon=0} \cdot \mathbf{S}_H = \begin{cases} 1, & \text{for horizontal polarisation } (\theta_0 = 0) \\ -1, & \text{for vertical polarisation } (\theta_0 = \frac{\pi}{2}) \end{cases} \quad (12)$$

shows that rotation error shortens the difference between the received Stokes vectors by a factor of $|\cos(2\epsilon)|$. The rotation impairment transforms the distinguishability as $d \rightarrow d|\cos(2\epsilon)|$. Modelling the rotation error as normally distributed $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$, equation 6 becomes

$$P_{\text{err}} \leq \mathbb{E}\left[e^{-C \cos^2(2\epsilon)}\right], \text{ where } C = \frac{d^2 N_S M}{k N_Z}. \quad (13)$$

Note that C is the original exponent in the Chernoff bound in equation 6 and can be loosely interpreted as the effective signal-to-noise ratio (SNR) considering the original thermal photon noise impairments. The factor of the $\cos^2(2\epsilon)$ is the additional rotational jitter impairment being proposed.

3) *Temporal Jitter Model*: In the ideal case, the signal and idler will arrive at the same time at the receiver to be jointly processed. However, in real systems there will be a small mismatch between the arrival times between the signal and idler photons. This can be caused by detector timing jitter, clock synchronisation errors between the TX and RX, and

dispersions in propagating lengths (e.g., moving platforms, vibration). To model these effects, the signal and idler are considered to be Gaussian pulses given by

$$f_{\text{signal}}(t) = f_{\text{idler}}(t) = \frac{1}{(2\pi\sigma_t^2)^{\frac{1}{4}}} \exp\left(\frac{-t^2}{4\sigma_t^2}\right). \quad (14)$$

where σ_t corresponds to the pulse width. The reason for writing the Gaussian in this form with $4\sigma_t^2$ on the denominator is to normalise the overlap factor; i.e., to get an overlap value of 1 for when there is no timing mismatch. The receiver expects $f_{\text{signal}}(t)$ to arrive at the receiver. However, due to timing jitter, $f_{\text{signal}}(t - \delta t)$ arrives, where $\delta t \sim \mathcal{N}(0, \sigma_j^2)$ is the timing mismatch between the signal and idler [13]. The overlap between the photon pair can be found using:

$$O(\delta t) = \int_{-\infty}^{\infty} f_{\text{idler}}(t) f_{\text{signal}}(t - \delta t) dt \quad (15)$$

$$= \exp\left(-\frac{\delta t^2}{8\sigma_t^2}\right). \quad (16)$$

We can see that $O(0) = 1$, and $O(\infty) = 0$, which makes sense. This overlap factor scales the distinguishability as $d \rightarrow d \cdot O(\delta t)$, such that less overlap reduces the distinguishability between quantum states at the receiver. With this in mind, the Chernoff bound becomes

$$P_{\text{err}} \leq \mathbb{E}_{\delta t} \left[\exp\left(-C e^{-\delta t^2/4\sigma_t^2}\right) \right]. \quad (17)$$

Here, C is again representing the baseline SNR with the thermal noise as the impairment, and the factor of $e^{-\delta t^2/4\sigma_t^2}$ is the proposed temporal jitter impairment.

C. Unified Impairment Model

If the effects of temporal, rotational, and pointing jitter are considered and independence is assumed, the three impairment factors from before combine to give $d \rightarrow d|\cos(2\epsilon)|e^{-\delta t^2/8\sigma_t^2}e^{-G\theta^2}$. Here, $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is the polarisation rotation error, $\delta t \sim \mathcal{N}(0, \sigma_j^2)$ is the temporal mismatch, and θ is the pointing error. This is different to Arnon's model where they use the pointing impairment factor as $\eta(\theta) \propto G^2 e^{-2G\theta^2}$ to capture the tradeoff where increasing the antenna gain increases signal power, but also increases pointing sensitivity for the optimisation problem. However, this work focuses on impairment-induced degradation, so the impairment model of $L(\theta) = e^{-G\theta^2}$ is used to derive the unified model. Since the Chernoff exponent depends on d^2 , the pointing factor contributes as $e^{-2G\theta^2}$. The pointing error θ is modelled as a Rayleigh distribution with probability density function $f(\theta) = \frac{\theta}{\sigma_p^2} e^{-\frac{\theta^2}{2\sigma_p^2}}$. Including these impairments, the Chernoff bound therefore becomes

$$P_{\text{err}} \leq \mathbb{E}_{\epsilon, \delta t, \theta} \left[\exp\left(-C \cos^2(2\epsilon) e^{-\delta t^2/4\sigma_t^2} e^{-2G\theta^2}\right) \right] \quad (18)$$

where C models the thermal-bath impairment. Note the triple expectation value over ϵ , δt , and θ , which will be evaluated in the next section.

V. NOVEL PERFORMANCE ANALYSIS

A. Pointing Jitter Model

The pointing jitter model from section IV-B1 is the following integral expression:

$$P_{\text{err}} \leq \alpha \int_0^\infty \frac{\theta}{\sigma_p^2} \exp\left(-\frac{\theta^2}{2\sigma_p^2} - k_1 G^2 e^{-2G\theta^2}\right) d\theta. \quad (19)$$

In [9], the integral was numerically solved, but a closed-form integral is derived here. To simplify the derivation, let $a_0 = \frac{\alpha}{\sigma_p^2}$, $a_1 = \frac{1}{2\sigma_p^2}$, $a_2 = k_1 G^2$, and $a_3 = 2G$. Then the expression becomes

$$P_{\text{err}} \leq \int_0^\infty a_0 \theta \exp(-a_1 \theta^2 - a_2 e^{-a_3 \theta^2}) d\theta \quad (20)$$

First, after the substitution $x = \theta^2$, the integral becomes

$$P_{\text{err}} \leq \frac{a_0}{2} \int_0^\infty [e^{-a_3 x}]^{\frac{a_1}{a_3}} \exp(-a_2 e^{-a_3 x}) dx. \quad (21)$$

Now the substitution $y = a_2 e^{-a_3 x}$ gives

$$P_{\text{err}} \leq \frac{a_0}{2a_3} a_2^{-\frac{a_1}{a_3}} \int_0^{a_2} y^{\frac{a_1}{a_3}-1} e^{-y} dy. \quad (22)$$

The incomplete gamma function is defined to be $\gamma(s, x) = \int_0^x t^{s-1} e^{-t} dt$. Inserting this and the constants a_0 , a_1 , a_2 , a_3 gives the closed-form expression:

$$P_{\text{err}} \leq \frac{\alpha}{4\sigma_p^2 G} (k_1 G^2)^{-\frac{1}{4\sigma_p^2 G}} \gamma\left(\frac{1}{4\sigma_p^2 G}, k_1 G^2\right). \quad (23)$$

Figure 3 is included to validate the derived result. It compares the closed-form expression against the direct numerical integration of Equation 7. The circular markers represent the numerical results obtained from quadrature evaluation of the integral, while the solid lines represent the corresponding values obtained from the derived incomplete-gamma closed form. The close agreement between the curves and markers across the full sweep of G confirms the correctness of the derived expression.

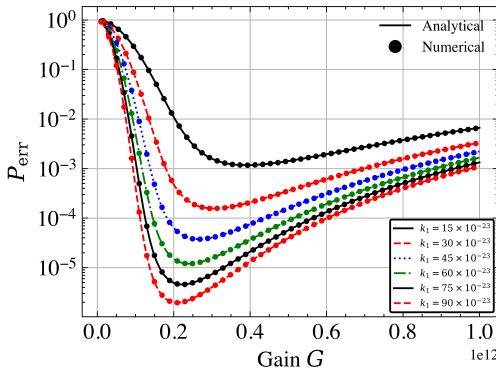


Fig. 3: Validation plot of P_{err} against G for various k values. The solid lines indicate analytical result and the circular markers show the numerical integration result

Prior work in [9] develops and evaluates the model via numerical integration, whereas the contribution here is the

derivation and validation of a closed-form expression in terms of the incomplete gamma function, which enables more efficient optimisation and parametric analysis for future study.

B. Rotation Jitter

In section IV-B2, the Chernoff bound for rotation error is

$$P_{\text{err}} \leq \mathbb{E}\left[e^{-C \cos^2(2\epsilon)}\right] \quad (24)$$

$$\leq e^{-\frac{C}{2}} \mathbb{E}\left[e^{-\frac{C}{2} \cos(4\epsilon)}\right] \quad (25)$$

where $C = \frac{d^2 N_S M}{k N_Z}$. We can reduce this expectation value using the Jacobi–Anger expansion [14] which transforms an expression in the form $e^{x \cos(\theta)}$ in terms of the modified Bessel function of the first kind $I_n(x)$. This expansion converts the nonlinear cosine term into a sum of harmonic components, which can be averaged term-by-term:

$$P_{\text{err}} \leq e^{-\frac{C}{2}} \mathbb{E}\left[I_0\left(\frac{C}{2}\right) + 2 \sum_{n=1}^{\infty} (-1)^n I_n\left(\frac{C}{2}\right) \cos(4n\epsilon)\right] \quad (26)$$

$$\leq e^{-\frac{C}{2}} \left(I_0\left(\frac{C}{2}\right) + 2 \sum_{n=1}^{\infty} (-1)^n I_n\left(\frac{C}{2}\right) \mathbb{E}[\cos(4n\epsilon)]\right). \quad (27)$$

The expectation value is found using

$$\mathbb{E}[\cos(4n\epsilon)] = \frac{1}{\sigma_\epsilon \sqrt{2\pi}} \int_{-\infty}^{\infty} \cos(4n\epsilon) e^{-\frac{\epsilon^2}{2\sigma_\epsilon^2}} d\epsilon. \quad (28)$$

Rewriting the cosine in terms of exponentials allows us to split the expression into two integral terms

$$\mathbb{E}[\cos(4n\epsilon)] = \frac{1}{2\sigma_\epsilon \sqrt{2\pi}} \left[\int_{-\infty}^{\infty} e^{-\frac{\epsilon^2}{2\sigma_\epsilon^2} + i4n\epsilon} d\epsilon + \int_{-\infty}^{\infty} e^{-\frac{\epsilon^2}{2\sigma_\epsilon^2} - i4n\epsilon} d\epsilon \right]. \quad (29)$$

This evaluates how random angular fluctuations attenuate each harmonic component. Completing the square allows us to write the integrals as shifted Gaussians:

$$\int_{-\infty}^{\infty} e^{-\frac{\epsilon^2}{2\sigma_\epsilon^2} \pm i4n\epsilon} d\epsilon = \int_{-\infty}^{\infty} \exp\left(-\frac{(\epsilon \mp i4n\sigma_\epsilon^2)^2}{2\sigma_\epsilon^2} - 8n^2\sigma_\epsilon^2\right) d\epsilon \quad (30)$$

$$= e^{-8n^2\sigma_\epsilon^2} \int_{-\infty}^{\infty} \exp\left(-\frac{(\epsilon \mp i4n\sigma_\epsilon^2)^2}{2\sigma_\epsilon^2}\right) d\epsilon. \quad (31)$$

This step isolates the oscillatory component and reveals a Gaussian decay factor in n . Finally, the standard Gaussian integral and Equation 31 can be used to rewrite Equation 29 as

$$\mathbb{E}[\cos(4n\epsilon)] = \frac{1}{2\sigma_\epsilon \sqrt{2\pi}} \left[e^{-8n^2\sigma_\epsilon^2} \sqrt{2\pi} \sigma_\epsilon + e^{-8n^2\sigma_\epsilon^2} \sqrt{2\pi} \sigma_\epsilon \right] \quad (32)$$

$$= e^{-8\sigma_\epsilon^2 n^2}. \quad (33)$$

The expectation value can be interpreted as the Fourier transform of a Gaussian, where the higher-frequency oscillations of the $\cos(4n\epsilon)$ term average out more rapidly, which leads to the exponential decay result. Inserting this, the expression for the Chernoff bound becomes

$$P_{\text{err}} \leq e^{-\frac{C}{2}} \left(I_0 \left(\frac{C}{2} \right) + 2 \sum_{n=1}^{\infty} (-1)^n I_n \left(\frac{C}{2} \right) e^{-8\sigma_{\epsilon}^2 n^2} \right). \quad (34)$$

This is the expression for P_{err} with the rotation jitter and thermal noise. It is now evident that rotation jitter introduces an exponential damping on higher-order terms, effectively reducing the distinguishability and increasing error probability.

C. Temporal Jitter

The Chernoff bound for temporal jitter was found in section IV-B3 to be

$$P_{\text{err}} \leq \mathbb{E}_{\delta t} \left[\exp \left(-C e^{-\delta t^2 / 4\sigma_t^2} \right) \right]. \quad (35)$$

where $C = \frac{d^2 N_S M}{k N_Z}$. Using the fact that δt is normally distributed,

$$P_{\text{err}} \leq \frac{1}{\sqrt{2\pi}\sigma_j} \int_{-\infty}^{\infty} \exp \left(-C e^{-\delta t^2 / 4\sigma_t^2} \right) \exp \left(-\frac{\delta t^2}{2\sigma_j^2} \right) d(\delta t). \quad (36)$$

Then rewriting the exponential as a power series:

$$\exp \left(-C e^{-\delta t^2 / 4\sigma_t^2} \right) = \sum_{m=0}^{\infty} \frac{\left(-C e^{-\delta t^2 / 4\sigma_t^2} \right)^m}{m!} \quad (37)$$

$$= \sum_{m=0}^{\infty} \frac{(-C)^m}{m!} e^{-m\delta t^2 / 4\sigma_t^2} \quad (38)$$

and inserting into Equation 36 gives

$$P_{\text{err}} \leq \frac{1}{\sqrt{2\pi}\sigma_j} \int_{-\infty}^{\infty} \sum_{m=0}^{\infty} \frac{(-C)^m}{m!} e^{-\frac{m\delta t^2}{4\sigma_t^2} - \frac{\delta t^2}{2\sigma_j^2}} d(\delta t) \quad (39)$$

$$\leq \frac{1}{\sqrt{2\pi}\sigma_j} \sum_{m=0}^{\infty} \frac{(-C)^m}{m!} \int_{-\infty}^{\infty} e^{-\frac{m\delta t^2}{4\sigma_t^2} - \frac{\delta t^2}{2\sigma_j^2}} d(\delta t), \quad (40)$$

where the interchange of summation and integration in the second step is justified by absolute integrability of the series, allowing the use of Fubini's theorem. To show the absolute integrability property, take the absolute value of the summand inside of the integral to give the series

$$S_{\text{integrand}} = \sum_{m=0}^{\infty} \frac{C^m}{m!} e^{-\frac{m\delta t^2}{4\sigma_t^2}} e^{-\frac{\delta t^2}{2\sigma_j^2}}. \quad (41)$$

Factor out the Gaussian term and the remaining sum becomes an exponential

$$S_{\text{integrand}} = e^{-\frac{\delta t^2}{2\sigma_j^2}} \sum_{m=0}^{\infty} \frac{\left(C e^{-\delta t^2 / (4\sigma_t^2)} \right)^m}{m!} \quad (42)$$

$$= e^{-\frac{\delta t^2}{2\sigma_j^2}} e^{C e^{-\delta t^2 / (4\sigma_t^2)}}. \quad (43)$$

Since $e^{-\delta t^2 / 4\sigma_t^2} \leq 1$, it can be written

$$S_{\text{integrand}} \leq e^C e^{-\frac{\delta t^2}{2\sigma_j^2}}. \quad (44)$$

This upper bound is a Gaussian function, which is integrable. Therefore, the series inside the integral is absolutely integrable, and the interchange of summation and integration is justified by Fubini's theorem. Finally, the Gaussian integral in Equation 40 is evaluated to give

$$P_{\text{err}} \leq \sum_{m=0}^{\infty} \frac{(-C)^m}{m!} \frac{1}{\sqrt{2\pi}\sigma_j} \sqrt{\frac{\pi}{\frac{m}{4\sigma_t^2} + \frac{1}{2\sigma_j^2}}} \quad (45)$$

$$\leq \sum_{m=0}^{\infty} \frac{(-C)^m}{m!} \left(\frac{m\sigma_j^2}{2\sigma_t^2} + 1 \right)^{-\frac{1}{2}}. \quad (46)$$

This is the analytical expression for the Chernoff bound considering the temporal jitter alongside the thermal noise contained within C .

It remains to show that this sum converges. Consider the absolute value series $S = \sum_{m=0}^{\infty} |a_m|$, where $|a_m| = \frac{C^m}{m!} \left(\frac{m\sigma_j^2}{2\sigma_t^2} + 1 \right)^{-\frac{1}{2}}$. Then,

$$\frac{|a_{m+1}|}{|a_m|} = \frac{C}{m+1} \sqrt{\frac{1 + \frac{m\sigma_j^2}{2\sigma_t^2}}{1 + \frac{(m+1)\sigma_j^2}{2\sigma_t^2}}} \quad (47)$$

$$< \frac{C}{m+1} \quad (48)$$

which is less than one for $m \geq \lceil C \rceil$ so the sum is absolutely convergent by the ratio test. For truncation, the original series is alternating, and for $m \geq \lceil C \rceil$ the terms decrease monotonically in magnitude. Therefore, for any truncation index $M \geq \lceil C \rceil$, the alternating-series remainder bound gives

$$\left| \sum_{m=M+1}^{\infty} a_m \right| \leq |a_{M+1}|. \quad (49)$$

Hence, to guarantee a truncation error below a prescribed threshold δ , it is sufficient to choose $M \geq \lceil C \rceil$ such that

$$|a_{M+1}| = \frac{C^{M+1}}{(M+1)!} \left(\frac{(M+1)\sigma_j^2}{2\sigma_t^2} + 1 \right)^{-1/2} \leq \delta. \quad (50)$$

Taking $C = 4$, $\sigma_j = 5$ ps, $\sigma_t = 20$ ps, and an error threshold of $\delta = 10^{-6}$, a numerical analysis gives $M = 20$ as the truncation length. These values were chosen to represent the moderate impairment regime.

D. Unified Model

The Chernoff bound for the combined temporal, rotational, and pointing jitter was found in IV-C to be

$$P_{\text{err}} \leq \mathbb{E}_{\epsilon, \delta t, \theta} \left[\exp \left(-C \cos^2(2\epsilon) e^{-\delta t^2 / 4\sigma_t^2} e^{-2G\theta^2} \right) \right], \quad (51)$$

The evaluation proceeds in three steps. First, the rotation expectation is evaluated using the Jacobi-Anger expansion, as before. Second, the temporal and pointing terms are kept

inside the exponent and expanded using the power-series forms of the exponential and modified Bessel functions. Finally, the temporal and pointing expectations are evaluated term-by-term, which produces an analytical series expression for P_{err} . Start by defining $A = Ce^{-\delta t^2/4\sigma_t^2}e^{-2G\theta^2}$. Then the unified bound can be written as

$$P_{\text{err}} \leq \mathbb{E}_{\epsilon, \delta t, \theta} \left[e^{-A \cos^2(2\epsilon)} \right]. \quad (52)$$

The cosine can be rewritten to give

$$e^{-A \cos^2(2\epsilon)} = e^{-A/2} e^{-(A/2) \cos(4\epsilon)}. \quad (53)$$

Applying the Jacobi-Anger expansion to the second exponential term gives

$$P_{\text{err}} \leq \mathbb{E}_{\delta t, \theta, \epsilon} \left[e^{-A/2} \left(I_0 \left(\frac{A}{2} \right) + 2 \sum_{n=1}^{\infty} (-1)^n I_n \left(\frac{A}{2} \right) \cos(4n\epsilon) \right) \right]. \quad (54)$$

Since A only depends on δt and θ , the expectation over ϵ can be evaluated separately using the result in equation 33 to give

$$P_{\text{err}} \leq \mathbb{E}_{\delta t, \theta} \left[e^{-A/2} \left(I_0 \left(\frac{A}{2} \right) + 2 \sum_{n=1}^{\infty} (-1)^n I_n \left(\frac{A}{2} \right) e^{-8n^2\sigma_\epsilon^2} \right) \right]. \quad (55)$$

To simplify the expression, introduce

$$c_n = \begin{cases} 1 & n = 0 \\ 2(-1)^n e^{-8n^2\sigma_\epsilon^2} & n \geq 1, \end{cases} \quad (56)$$

which gives

$$P_{\text{err}} \leq \mathbb{E}_{\delta t, \theta} \left[\sum_{n=0}^{\infty} c_n e^{-A/2} I_n \left(\frac{A}{2} \right) \right]. \quad (57)$$

Next, the remaining two expectations are evaluated by expanding the two functions containing A . Using

$$e^{-x} = \sum_{m=0}^{\infty} \frac{(-x)^m}{m!}, \quad I_n(x) = \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(k+n+1)} \left(\frac{x}{2} \right)^{2k+n}, \quad (58)$$

the functions become

$$e^{-A/2} I_n \left(\frac{A}{2} \right) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^m}{m! k! \Gamma(k+n+1)} \left(\frac{A}{2} \right)^m \left(\frac{A}{4} \right)^{2k+n}. \quad (59)$$

Substituting the expression for A back in gives

$$e^{-A/2} I_n \left(\frac{A}{2} \right) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^m (C/2)^m (C/4)^{2k+n}}{m! k! \Gamma(k+n+1)} u^v w^v, \quad (60)$$

where $v = m + 2k + n$, $u = e^{-\delta t^2/4\sigma_t^2}$, $w = e^{-2G\theta^2}$. Since δt and θ are assumed to be independent, their expectations factorise as $\mathbb{E}_{\delta t, \theta} [u^v w^v] = \mathbb{E}_{\delta t} [u^v] \mathbb{E}_{\theta} [w^v]$, which means we can evaluate the expectation value of the temporal and pointing

impairments separately. Using this and substituting equation 60 into equation 57 gives

$$P_{\text{err}} \leq \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{c_n (-1)^m (C/2)^m (C/4)^{2k+n}}{m! k! \Gamma(k+n+1)} \mathbb{E}_{\delta t} [u^v] \mathbb{E}_{\theta} [w^v]. \quad (61)$$

The temporal expectation can be written using the normal probability density function as

$$\begin{aligned} \mathbb{E}_{\delta t} [u^v] &= \frac{1}{\sqrt{2\pi}\sigma_j} \int_{-\infty}^{\infty} \exp \left[- \left(\frac{1}{2\sigma_j^2} + \frac{v}{4\sigma_t^2} \right) \delta t^2 \right] d(\delta t) \\ &= \frac{1}{\sqrt{1 + \frac{v\sigma_j^2}{2\sigma_t^2}}} \end{aligned} \quad (62)$$

where the second step comes from the standard Gaussian integral. The pointing expectation can be written using the Rayleigh probability distribution as

$$\mathbb{E}_{\theta} [w^v] = \frac{1}{\sigma_p^2} \int_0^{\infty} \theta \exp \left[- \left(\frac{1}{2\sigma_p^2} + 2Gv \right) \theta^2 \right] d\theta. \quad (63)$$

This integral can be solved by considering the integral

$$\int_0^{\infty} \theta e^{-a\theta^2} d\theta = \frac{1}{2a}, \quad (64)$$

which was evaluated by the substitution $u = \theta^2$. Setting $a = \frac{1}{2\sigma_p^2} + 2Gv$ recovers equation 63, this means

$$\mathbb{E}_{\theta} [w^v] = \frac{1}{\sigma_p^2} \cdot \frac{1}{2 \left(\frac{1}{2\sigma_p^2} + 2Gv \right)} \quad (65)$$

$$= \frac{1}{1 + 4G\sigma_p^2 v}. \quad (66)$$

This shows the pointing jitter induces a rational attenuation factor rather than exponential decay after averaging. Now that the two expectations are properly evaluated, they can finally be inserted back into equation 61 to give

$$\begin{aligned} P_{\text{err}} &\leq \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{c_n (-1)^m (C/2)^m (C/4)^{2k+n}}{m! k! \Gamma(k+n+1)} \\ &\quad \times \frac{1}{\sqrt{1 + \frac{v\sigma_j^2}{2\sigma_t^2}}} \frac{1}{1 + 4G\sigma_p^2 v}. \end{aligned} \quad (67)$$

This is the analytical expression for P_{err} considering the thermal-bath, pointing, rotation, and temporal impairments. All that is left to show is that the series converges. The absolute convergence condition is $\sum_n \sum_m \sum_k |a_{n,m,k}| < \infty$, with

$$a_{n,m,k} = \frac{c_n (-1)^m (C/4)^{2k+n} (C/2)^m}{m! k! \Gamma(k+n+1) \sqrt{1 + \frac{v\sigma_j^2}{2\sigma_t^2}} (1 + 4G\sigma_p^2 v)}. \quad (68)$$

First note that $\sqrt{1 + \frac{v\sigma_j^2}{2\sigma_t^2}} \geq 1$ and $1 + 4G\sigma_p^2 v \geq 1$ so these factors can be removed. Also note that $|c_n| \leq 2$, and $\Gamma(k + n + 1) = (k + n)! \geq k!n!$. Hence

$$|a_{n,m,k}| \leq 2 \frac{(C/4)^{2k+n} (C/2)^m}{m!k!n!} \quad (69)$$

$$\leq 2 \frac{(C/4)^{2k+n} (C/2)^m}{m!k!n!} \quad (70)$$

$$\leq 2 \frac{(C/2)^m}{m!} \cdot \frac{(C^2/16)^k}{k!} \cdot \frac{(C/4)^n}{n!}. \quad (71)$$

Therefore,

$$\sum_{n,m,k} |a_{n,m,k}| \leq 2 \left(\sum_{m=0}^{\infty} \frac{(C/2)^m}{m!} \right) \left(\sum_{k=0}^{\infty} \frac{(C^2/16)^k}{k!} \right) \left(\sum_{n=0}^{\infty} \frac{(C/4)^n}{n!} \right) \quad (72)$$

$$\leq 2 e^{C/2} e^{C^2/16} e^{C/4} \quad (73)$$

which is clearly finite, so the series converges absolutely and Equation 67 is a closed-form expression. To find adequate truncation values, numerical methods were used to find truncation lengths for each of the three summations such that the absolute error was below some threshold δ . That is, $|\sum_{n,m,k}^{\infty} a_{n,m,k} - \sum_{n,m,k}^{N,M,K} a_{n,m,k}| < \delta$.

For $C = 4$, a truncation length of $N = 7$, $M = 12$, and $K = 5$ is adequate for purposes where an error of 10^{-6} is acceptable. Higher C demands more terms to be included and hence values of N , M and K .

E. Model Summary

Below is the table summarising the channel models and their assumptions.

TABLE I: Summary of channel models and analytical expressions. All models are constructed on top of the thermal-bath model.

Model (Impairments)	Assumptions	Eq.
Pointing (θ)	θ Rayleigh; FSO model	(23)
Rotation (ϵ)	$\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$; Stokes model	(34)
Temporal (δt)	$\delta t \sim \mathcal{N}(0, \sigma_j^2)$; Gaussian pulses	(46)
Unified ($\theta, \epsilon, \delta t$)	independent contraction	(67)

VI. BER OPTIMISATION DEFINITION AND SOLUTION METHODOLOGY

A. Optimisation Formulation

The optimisation problem is to minimise the error probability expression for the unified model,

$$\min_{N_S, \sigma_t} P_{\text{err}}(N_S, \sigma_t) \quad (74)$$

which comes from Equation 51:

$$P_{\text{err}}(N_S, \sigma_t) \leq \mathbb{E}_{\delta t, \epsilon, \theta} \left[\exp \left(-D N_S e^{-\delta t^2/4\sigma_t^2} \right) \right], \quad (75)$$

where $D = \frac{d^2 M}{k N_Z} \cos^2(2\epsilon) e^{-2G\theta^2}$ is introduced to group the factors that are independent of N_S and σ_t . Here, the

design parameters are N_S and σ_t and everything else is fixed by the system. The optimisation is subject to the following constraints:

1) Resource constraint

$$N_S \sigma_t \leq B \quad (76)$$

for some constant bound B . This couples signal brightness and pulse duration. Higher signal brightness and longer pulses both increase resource cost, so the transmitter cannot support arbitrarily large values of both simultaneously.

2) Performance constraint

$$\sigma_t \leq \frac{1}{\kappa R_{\min}} \quad (77)$$

which captures the inverse relationship between the signal duration and the achievable data rate $R_{\min}$ for some proportionality constant κ .

3) Feasibility constraint

$$N_S > 0 \quad \sigma_t \geq \sigma_t^{\min} \quad (78)$$

which ensures the system remains physically valid.

B. Equivalent Transformations and Solution Approach

At the optimum point, the full resource budget B is utilised. For fixed σ_t , the objective is monotonically decreasing in N_S since it appears only in the negative exponent, so any feasible point with $N_S \sigma_t < B$ can be improved by increasing N_S . Therefore, the resource constraint is active at optimum, giving the equality

$$N_S = \frac{B}{\sigma_t}. \quad (79)$$

Substituting into the objective reduces it to a one-dimensional optimisation:

$$\min_{\sigma_t} \mathbb{E}_{\delta t, \epsilon, \theta} \left[\exp \left(-D \frac{B}{\sigma_t} e^{-\delta t^2/4\sigma_t^2} \right) \right]. \quad (80)$$

This dimension reduction is exact under the assumption that the resource constraint is active at optimum ($N_S \sigma_t = B$), since for fixed σ_t the objective is monotonically decreasing in N_S , ensuring optimality is preserved. The range of the pulse width is subject to

$$\sigma_t^{\min} \leq \sigma_t \leq \frac{1}{\kappa R_{\min}}. \quad (81)$$

This transformation reveals an important trade off. Small σ_t increases the data rate and the signal brightness ($N_S = B/\sigma_t$) but reduces the temporal overlap, degrading the quantum correlation. Large σ_t increases overlap and robustness to temporal jitter, but reduces data rate and signal brightness. Since the rotation and temporal impairments are Gaussian and the pointing impairment is Rayleigh distributed, Gauss-Hermite quadrature is used for the Gaussian variables and Gauss-Laguerre quadrature is used for the Rayleigh pointing variable [14]. This converts the continuous integrals into finite weighted sums that can be efficiently evaluated for each σ_t . This approach

provides high accuracy with relatively low computational cost compared to Monte Carlo methods, with accuracy controlled by the quadrature order. The resulting scalar objective is then minimised using bounded numerical optimisation based on Brent's method [15]. The baseline parameters used in the optimisation are as follows: $B = 0.5$ ps, $d = 1$, $M = 10^4$, $\kappa = 1$, $N_Z = 20$, $\sigma_\epsilon = 7.5^\circ$, $\sigma_j = 5$ ps, $\sigma_t^{\min} = 1$ ps, $R_{\min} = 0.05$ ps $^{-1}$, $G = 10^6$, and $\sigma_p = 1.3 \times 10^{-4}$ rad. These values were chosen to represent a moderate-impairment regime. The resource bound B was chosen as 0.5 ps to ensure that the low signal brightness condition was still satisfied to achieve the quantum advantage. The minimum data rate $R_{\min}$ was chosen as 0.05 ps $^{-1}$ to prevent excessively long pulses that would otherwise reduce error probability at the expense of sacrificing throughput. Using these values, multiple plots were made for various values of the resource budget B in Figure 4.

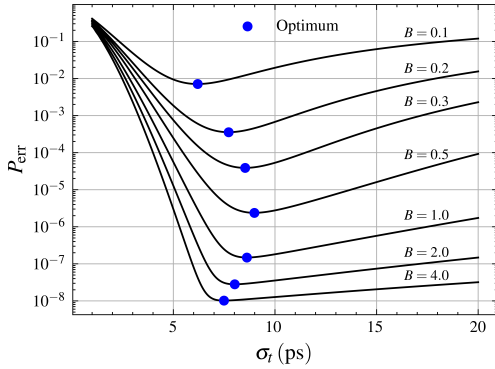


Fig. 4: Bit error probability versus pulse width σ_t for various values of resource budget B . The marked point indicates the numerically obtained optimum.

Therefore, to achieve the optimum performance under the constraints considered here, the values of σ_t^* and $N_S^* = B/\sigma_t^*$ should be set at the transmitter for the relevant value of B to ensure minimum P_{err} .

Figure 5 is included to benchmark the optimised performance by comparing four cases: the fully optimised solution $\sigma_t^*(B)$, the minimum pulse width $\sigma_{t,\min}$, the maximum pulse width set by the data-rate constraint $\sigma_{t,\max} = 1/(\kappa R_{\min})$, and a nominal fixed pulse width obtained by the geometric mean $\sigma_{t,\text{nominal}} = \sqrt{\sigma_{t,\min}\sigma_{t,\max}}$. For each budget value B , the optimised curve is obtained by solving the constrained P_{err} minimisation problem for each B point. The other three benchmark curves are found by holding σ_t fixed at these representative values and computing the corresponding P_{err} .

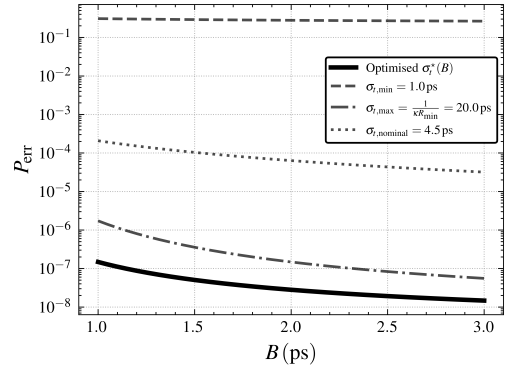


Fig. 5: Optimisation benchmark plot showing the performance of the minimum, maximum, nominal and optimised pulse width.

From the figure, the optimised curve provides the best performance as expected. The other three curves perform worse because they cannot balance the competing effects of signal brightness N_S and temporal robustness. In particular, small σ_t values tend to favour higher signal brightness $N_S = B/\sigma_t$ but suffer from stronger temporal-jitter sensitivity, whereas large σ_t values improve jitter robustness but reduce the achievable signal brightness due to the resource budget B .

C. Complexity Analysis

The computational complexity is dominated by quadrature evaluation. For quadrature order $Q = 200$, each integral is converted to a discrete sum of Q terms. Therefore, the two Gaussian integrals and one Rayleigh integral are converted into a three-dimensional quadrature sum with a total of $Q^3 = 8 \times 10^6$ terms, so the Gaussian-Hermite/Gauss-Laguerre approximation requires $O(Q^3)$ computations per evaluation of the objective function. If the optimisation routine requires $K = 12$ objective function evaluations ($\sigma_t^1, \sigma_t^2, \dots, \sigma_t^K$), then the overall complexity is $O(KQ^3) = 9.6 \times 10^7$ operations. This remains feasible since the optimisation problem was reduced to a one-dimensional search. Lower-complexity alternatives can be achieved by: (i) Reducing Q and/or K . (ii) Using the small-jitter approximation allows the impairments to be expanded as $\cos(2\epsilon) \approx 1 - 2\epsilon^2$ and $e^{-\delta t^2/(4\sigma_t^2)} \approx 1 - \delta t^2/(4\sigma_t^2)$, where the expectations can now be evaluated as simple Gaussian moments. (iii) Truncating the closed-form series in Equation 67 to avoid the full three-dimensional quadrature.

VII. NUMERICAL RESULTS AND PERFORMANCE EVALUATION

A. Default Parameters

Unless otherwise stated, the following values are used to simulate the models.

TABLE II: Default simulation parameters showing the baseline values used in the plots.

Parameter	Symbol	Value
Effective SNR parameter	$C = \frac{d^2 N_S M}{k N_Z}$	4
Rotation jitter std. dev.	σ_ϵ	7.5°
Pulse width	σ_t	20 ps
Timing jitter std. dev.	σ_j	5 ps
Relative timing jitter	σ_j/σ_t	0.25
Pointing jitter std. dev.	σ_p	0.0075°
Pointing gain parameter	G	10^6
Monte Carlo samples	N_{MC}	10^6
Series truncation tolerance	ϵ_{tol}	10^{-10}

B. Validation and Design Insights

The unified model is validated by comparing the derived analytical expressions for P_{err} against Monte Carlo simulations for several fixed values of C . In each of the plots, P_{err} was obtained by varying one impairment parameter and holding the other two fixed. The solid curves correspond to the analytical results, and the dashed red curves correspond to Monte Carlo estimates obtained using 10^5 samples per parameter value. For each case, the simulation samples the Gaussian rotation and temporal variables, and the Rayleigh pointing variable directly before computing the mean for these samples in Equation 51. The overlap between analytical and simulated results verifies the accuracy of the derived expressions.

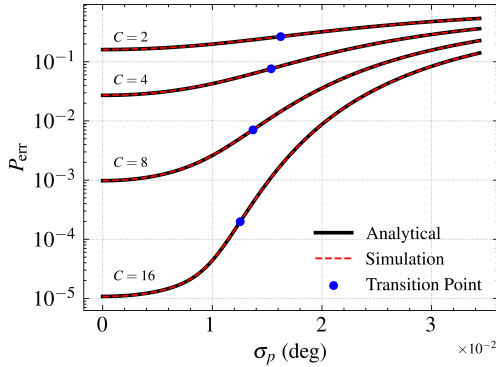


Fig. 6: Error probability P_{err} versus pointing jitter σ_p for the combined impairment model over various values of C , with σ_j and σ_ϵ held constant at their baseline value. This figure is included for validation and to visualise the effect of the pointing impairment.

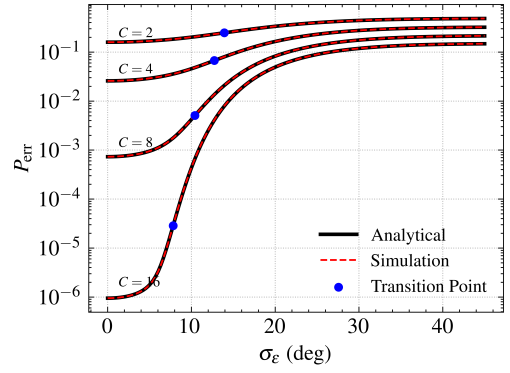


Fig. 7: Error probability P_{err} versus rotation jitter σ_ϵ for the combined impairment model over various values of C , with σ_j and σ_p held constant at their baseline value. This figure is included for validation and to visualise the effect of the rotation impairment.

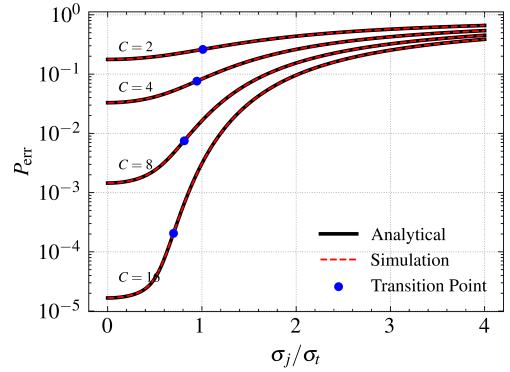


Fig. 8: Error probability P_{err} versus the dimensionless temporal jitter ratio σ_j/σ_t for the combined impairment model over various values of C , with σ_ϵ and σ_p held constant at their baseline value. This figure is included for validation and to visualise the effect of the temporal impairment.

In all three plots, P_{err} decreases as we increase C since we have more signal photons per thermal noise photon. In real systems, the impairment severity σ_p , σ_ϵ , and σ_j are not constant due to environmental changes. Therefore, robust system design should aim to minimise the changes in P_{err} in real environments. All figures exhibit a transition from concave-up to concave-down behaviour, indicating three operating regimes: an initial robustness region, a rapid degradation region, and a saturation region. The main design implication is that the system should be designed to operate in the robust region, approximately $0^\circ \leq \sigma_p \leq 0.008^\circ$ and $0^\circ \leq \sigma_\epsilon \leq 6^\circ$ and $0 \leq \sigma_j/\sigma_t \leq 0.4$. This means environmental changes have little effect on P_{err} since the curve in that region is relatively flat.

C. Performance Comparisons

To compare the models, Figure 9 is included which plots P_{err} against the effective SNR parameter C for five cases: the

idealised thermal-bath model, the pointing jitter, the rotation-jitter, the temporal-jitter and thermal-bath model, and the full model with thermal noise and all three practical impairments. The thermal-bath model is used in each. The values of C are chosen to be around the baseline value $C = 4$.

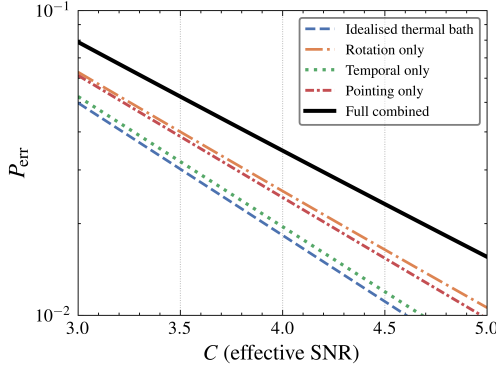


Fig. 9: Benchmark for impairment models around the baseline value of $C = 4$. This provides a direct comparison between the proposed models in this work and the standard models in literature.

The idealised thermal-bath curve is used as the reference and gives the lowest P_{err} since it assumes no additional impairments. The temporal-jitter remains close to thermal impairment curve. The pointing-only curve lies close to the rotation-only curve, while both produce substantially larger P_{err} than the temporal-only case.

This was further investigated using sensitivity analysis. For the baseline $P_{\text{err}} = 0.114$, the disturbance required to achieve a twofold increase in error probability was computed for each impairment.

TABLE III: Sensitivity analysis showing multiplier required to double P_{err} .

Impairment	Multiplier	Final Value	Relative Increase (%)
Pointing	2.54	0.019°	153.9
Rotation	2.33	17.50°	133.3
Temporal	4.99	1.248	399.0

It was found that pointing and rotation jitter requires only a 2.54 and 2.33 fold increase in severity to double P_{err} , whereas temporal jitter requires a significantly larger 4.99 fold increase. These results demonstrate that rotation jitter is the dominant local impairment, followed by pointing jitter, while temporal jitter is less sensitive around the selected baseline. From a design perspective, this suggests that tighter control of polarisation and antenna pointing alignment should be prioritised over timing synchronisation.

D. Experimental Implementation

The proposed models are compatible with existing quantum illumination apparatus and are intended as a conceptual feasibility discussion rather than a detailed hardware implementation. A practical realisation would require a source of

entangled photon pairs (e.g., spontaneous parametric down-conversion [16] based TMSV source), polarisation modulators at the tag, and a joint receiver such as a parametric amplifier [17]. Quantum receivers that are unable to distinguish polarisation states should be accompanied by a polarising beamsplitter in order to separate polarisation states into spatial states. Additionally, delay lines and beam steering devices are needed to control temporal and pointing jitter. The main experimental challenges include maintaining stable signal-idler entanglement over the round-trip path, achieving timing synchronisation at the picosecond scale, and ensuring precise polarisation and pointing alignment. These constraints directly correspond to the impairment parameters modelled in this work, supporting the relevance of the proposed BER framework for realistic system design.

VIII. CONCLUDING REMARKS AND FUTURE DIRECTIONS

This work developed analytical performance models for QBC under practical impairments, including rotation, temporal and pointing jitter. Closed-form expressions for P_{err} were derived and validated, showing how these impairments contract the effective distinguishability and degrade detection performance beyond the ideal thermal-bath models. Rigorous convergence arguments were provided to ensure the correctness of the derivation. A key contribution is the development of a unified impairment framework that captures pointing, rotation, and temporal jitter within a single analytical model, enabling direct evaluation of their combined impact on system performance. This improves upon prior work that treats impairments in isolation.

The results demonstrate that practical impairments introduce nonlinear degradation effects that are not captured by existing models, and that system performance is highly sensitive to rotational and pointing jitter in the considered regime. Robust operating regimes were identified through parameter sweeps of the impairments which mark regions of low P_{err} fluctuations suitable for system design. Furthermore, the proposed optimisation framework reveals a fundamental trade-off between pulse width and signal brightness, which enables the identification of operating points that balance temporal robustness (large σ_t) and resource efficiency (B). These findings provide actionable design insight, particularly in prioritising polarisation stability and pointing alignment over timing precision for improved system performance.

Future work will extend the framework to multi-tag QBC scenarios, including joint optimisation of resource allocation and interference management across multiple tags. Additional impairment effects, such as idler loss, detector inefficiency, and entangled-state impurity, will be included to further bridge theory and realistic implementations. Experimental validation will focus on verifying the accuracy of the proposed BER models by comparing error rates against analytical expressions under controlled impairments. The main challenges include maintaining entanglement over the round-trip channel, and isolating individual impairment effects to enable meaningful

model validation, thereby assessing the accuracy of the proposed framework in practical systems.

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